Patch-Based Diffusion Models for Solving Inverse Problems in Computed Tomography Imaging

Executive Summary

Thomas Hartigan | MPhil in Data Intensive Science

Word count: 999

Purpose and Motivation

Computed tomography (CT) reconstructs internal structure from X-ray projections. Acquiring fewer projections can reduce scan time and radiation exposure. However, sparse angular sampling makes the reconstruction problem underdetermined, causing streak artefacts and loss of fine structure without an image prior [1].

Diffusion models provide such priors by learning how noisy images should be updated towards the data distribution [2]. Whole-image models retain global anatomical context, but their memory requirements increase with resolution. In contrast, the Patch Diffusion Inverse Solver (PaDIS) trains on position-labelled patches and combines their scores across randomly shifted layouts [3]. Hu et al. reported that PaDIS reduced training cost and consistently outperformed a comparable whole-image model on sparse-view CT.

In this work, I test whether this claim generalises to public chest CT data and a fully specified fan-beam operator. I also investigate which implementation choices determine the comparison and introduce operator-derived scaling for measurement updates. This separates scientific reproduction from the engineering changes required for a new CT geometry.

Approach

To separate ambiguities in the source material from transfer to a new CT problem, I implemented three tracks. First, Paper-Faithful follows the published algorithms. Repo-Faithful then follows the released code [4]. Across three matched reconstructions, an adapted version achieved cross-implementation structural similarity of 0.996. Finally, LION-Physical replaces geometry-specific measurement scales with values derived from the CT operator [5], while retaining the same learned priors.

Training, tuning, and evaluation were implemented in LION [6] using patient-separated images from the Lung Image Database Consortium and Image Database Resource Initiative (LIDC-IDRI) [7]. Tomosipo and ASTRA generated noise-free, two-dimensional fan-beam measurements with specified physical dimensions [8, 9].

Main experiments reconstruct 256×256 images from 8 or 20 views over a full rotation. Further tests use 60 views, limited angles, and native 512×512 resolution. Comparators include Feldkamp–Davis–Kress filtered back-projection (FDK) [10], Chambolle–Pock total-variation reconstruction (CP) [11, 12], and whole-image diffusion. Diffusion experiments use several samplers, including variance-exploding diffusion posterior sampling (VE-DPS). Peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) measure image quality [13], while relative projection residuals measure consistency with the simulated data. Figures use Hounsfield-unit (HU) and normalised-intensity (NI) displays.

Principal Findings

Figure 1 shows that both learned priors reduce sparse-view artefacts relative to FDK and CP. However, the matched LION-Physical VE-DPS comparison [14] favoured whole-image diffusion. At 20 views, whole-image and PaDIS PSNRs were $34.25^{+0.71}_{-0.66}$ and $32.82^{+1.06}_{-1.06}$ dB, respectively. At 8 views, they were $29.58^{+0.68}_{-0.60}$ and $27.65^{+0.75}_{-0.77}$ dB. Paper-Faithful results gave the same ordering. This suggests that global context remains beneficial when whole-image training is feasible.

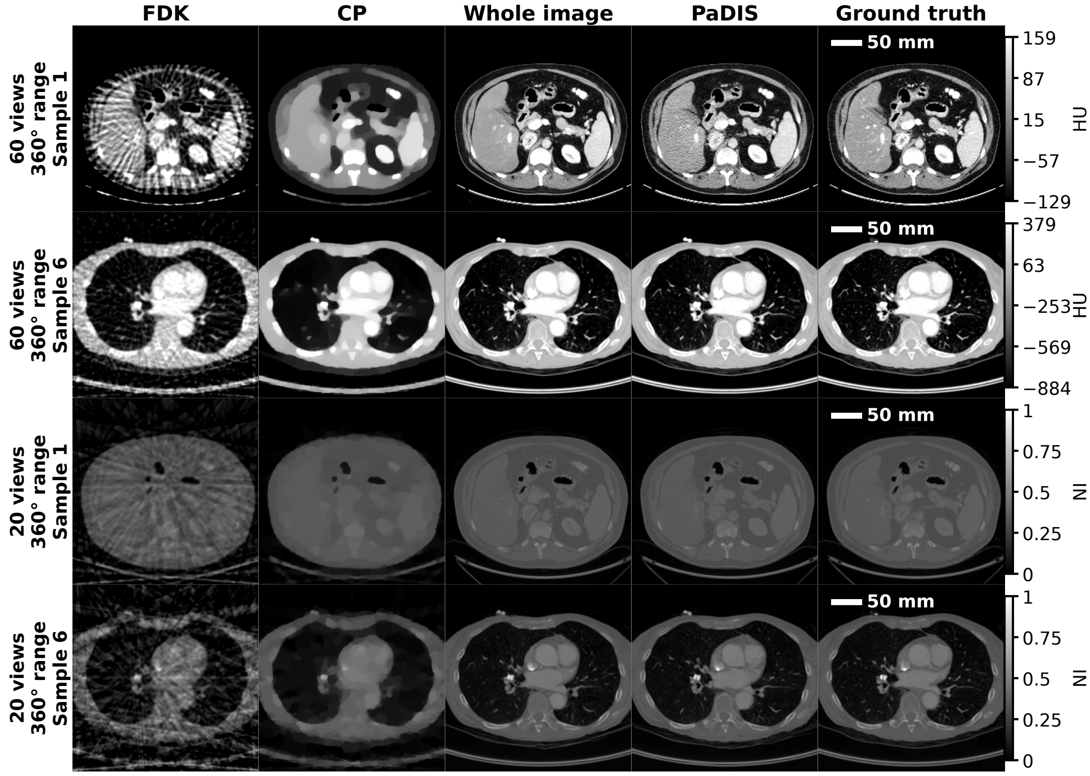

Figure 1: Representative 60-view HU and 20-view NI reconstructions. Whole image and PaDIS denote the corresponding priors used with VE-DPS. Both learned methods reduce visible sparse-view artefacts relative to FDK and CP.

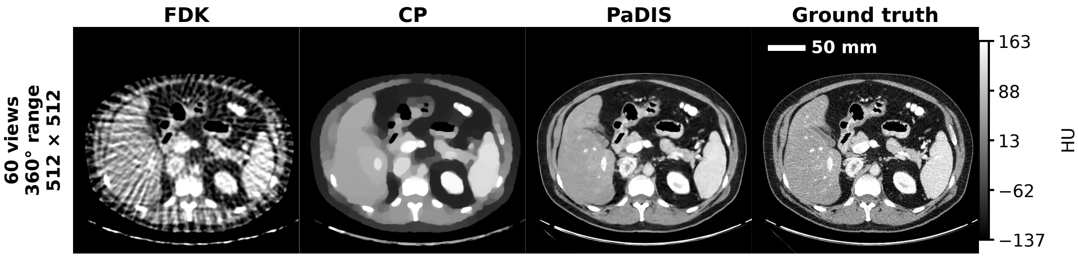

Figure 2: Native 512×512 reconstruction from 60 views in HU. PaDIS-VE-DPS preserves more anatomical detail than FDK and CP without requiring a whole-image model at this resolution.

In contrast, PaDIS remains applicable when memory constraints prevent whole-image training. The four-image native-resolution experiment in Figure 2 achieved $40.08^{+0.68}_{-1.11}$ dB PSNR and $0.960^{+0.004}_{-0.004}$ SSIM at 512×512, exceeding CP by 6.31 dB. However, no whole-image model was trained at this resolution, and the small sample prevents a population-level comparison. Therefore, this result demonstrates high-resolution feasibility rather than superiority over whole-image diffusion.

Sampler choice changed the ranking further. With the separately tuned variance-exploding denoising diffusion null-space model (VE-DDNM) [15], PaDIS achieved the highest PSNR at 20 and 60 views, reaching $36.49^{+0.95}_{-0.94}$ and $41.55^{+1.38}_{-1.52}$ dB. However, each prior–sampler pair was tuned separately, so the comparison does not isolate prior architecture. These results show that PaDIS performance depends on how its patch score is used.

LION-Physical VE-DPS also reduced relative projection residuals to approximately 10^{-3} across several geometries, over an order of magnitude below many reproduction-track results. This indicates that operator-derived scaling produces reconstructions closely supported by the measurements. A low residual does not, however, establish anatomical accuracy because sparse-view operators leave poorly constrained image components.

Ablations show that patch size and noise schedule affect reconstruction more than positional channels or initialisation. Under a shared inference setting, the 96×96 patch model reached 34.26 dB, compared with 34.25 dB for whole-image diffusion, while smaller patches gave no consistent benefit. Removing positional channels reduced PSNR by only 0.3–0.4 dB, whereas the geometric schedule improved it from 31.53 to 32.82 dB. However, non-default patch models were not tuned separately. Under fixed training time, using all processed slices reduced PaDIS PSNR to 31.68 dB and exposure per image. Therefore, this does not show that additional data are harmful.

Finally, Figure 3 shows that randomly shifted PaDIS layouts suppress the discontinuities produced by fixed stitching and averaging. Whole-image samples were nevertheless more anatomically credible. As only four samples were generated per method, this suggests a continuity benefit but cannot rank distributional quality.

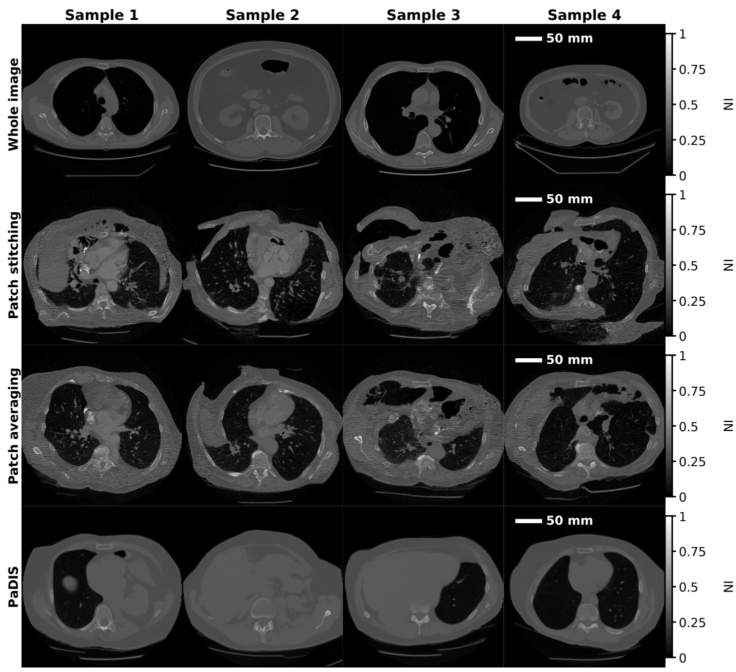

Figure 3: Four unconditional NI samples per prior. Fixed patch layouts introduce discontinuities, while PaDIS improves continuity but remains less anatomically credible than whole-image diffusion.

Reproducibility, Limitations, and Implications

Reproduction identified material gaps between the original paper and released code. Architecture, validation, checkpoint selection, physical geometry, tuning, and sampling are incompletely or inconsistently specified [3, 4]. The three implementation tracks show that plausible interpretations can reverse the central comparison. Therefore, complete geometry, data splits, tuning procedures, and executable algorithms are required to distinguish methodological improvements from implementation effects.

Several limitations constrain generalisation. Experiments use one dataset, simulated two-dimensional measurements, and one training run per model. Most results cover 25 slices rather than independent scans, while expensive comparisons use four images. Furthermore, photon noise, scatter, beam hardening, and motion are excluded. Reported intervals describe variation between slices rather than retraining uncertainty, and no clinical expert assessed lesion preservation or hallucination. Clinical low-dose performance has therefore not been established.

Overall, PaDIS provides a memory-scalable route to high-resolution diffusion priors, but it does not consistently outperform whole-image diffusion. LION-Physical provides operator-derived data-consistency scaling, while sampler-specific validation remains necessary. Future work could investigate calibrated measurement noise, three-dimensional operators, and multi-seed training under equal-epoch and equal-compute budgets. Evaluation on independent scans with clinically relevant tasks and expert assessment would then test whether improved image metrics correspond to reliable clinical structure. Project code is available here, with training runs and saved models available here.

References

[1] E. Y. Sidky, C.-M. Kao, and X. Pan. "Accurate Image Reconstruction from Few-Views and Limited-Angle Data in Divergent-Beam CT". In: *Journal of X-Ray Science and Technology* 14.2 (May 2006), pp. 119–139. ISSN: 0895-3996. DOI: 10.3233/XST-2006-00155. (Visited on 07/12/2026).

[2] Y. Song et al. "Score-Based Generative Modeling through Stochastic Differential Equations". In: *International Conference on Learning Representations*. Oct. 2020. (Visited on 07/12/2026).

[3] J. Hu et al. "Learning Image Priors Through Patch-Based Diffusion Models for Solving Inverse Problems". In: *Advances in Neural Information Processing Systems*. Vol. 37. Curran Associates, Inc., 2024, pp. 1625–1660. DOI: 10.52202/079017-0052. (Visited on 07/12/2026).

[4] J. Hu. *Jasonhu4/PaDIS*. Mar. 2026. (Visited on 07/12/2026).

[5] A. Beck and M. Teboulle. "A Fast Iterative Shrinkage-Thresholding Algorithm for Linear Inverse Problems". In: *SIAM Journal on Imaging Sciences* 2.1 (Jan. 2009), pp. 183–202. ISSN: 1936-4954. DOI: 10.1137/080716542. (Visited on 07/12/2026).

[6] M. B. Kiss et al. "Benchmarking Learned Algorithms for Computed Tomography Image Reconstruction Tasks". In: *Applied Mathematics for Modern Challenges* 3.0 (Feb. 2025), pp. 1–43. DOI: 10.3934/ammc.2025001. (Visited on 07/12/2026).

[7] S. G. Armato et al. "The Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI): A Completed Reference Database of Lung Nodules on CT Scans". In: *Medical Physics* 38.2 (Feb. 2011), pp. 915–931. ISSN: 0094-2405. DOI: 10.1118/1.3528204.

[8] A. A. Hendriksen et al. "Tomosipo: Fast, Flexible, and Convenient 3D Tomography for Complex Scanning Geometries in Python". In: *Optics Express* 29.24 (Nov. 2021), p. 40494. ISSN: 1094-4087. DOI: 10.1364/OE.439909. (Visited on 07/12/2026).

[9] W. Van Aarle et al. "Fast and Flexible X-ray Tomography Using the ASTRA Toolbox". In: *Optics Express* 24.22 (Oct. 2016), p. 25129. ISSN: 1094-4087. DOI: 10.1364/OE.24.025129. (Visited on 07/12/2026).

[10] L. A. Feldkamp, L. C. Davis, and J. W. Kress. "Practical Cone-Beam Algorithm". In: *JOSA A* 1.6 (June 1984), pp. 612–619. ISSN: 1520-8532. DOI: 10.1364/JOSAA.1.000612. (Visited on 07/12/2026).

[11] L. I. Rudin, S. Osher, and E. Fatemi. "Nonlinear Total Variation Based Noise Removal Algorithms". In: *Physica D: Nonlinear Phenomena* 60.1-4 (Nov. 1992), pp. 259–268. ISSN: 01672789. DOI: 10.1016/0167-2789(92)90242-F. (Visited on 07/12/2026).

[12] A. Chambolle and T. Pock. "A First-Order Primal-Dual Algorithm for Convex Problems with Applications to Imaging". In: *Journal of Mathematical Imaging and Vision* 40.1 (May 2011), pp. 120–145. ISSN: 0924-9907, 1573-7683. DOI: 10.1007/s10851-010-0251-1. (Visited on 07/12/2026).

[13] Z. Wang et al. "Image Quality Assessment: From Error Visibility to Structural Similarity". In: *IEEE Transactions on Image Processing* 13.4 (Apr. 2004), pp. 600–612. ISSN: 1941-0042. DOI: 10.1109/TIP.2003.819861. (Visited on 07/12/2026).

[14] H. Chung et al. "Diffusion Posterior Sampling for General Noisy Inverse Problems". In: *The Eleventh International Conference on Learning Representations*. Sept. 2022. (Visited on 07/12/2026).

[15] Y. Wang, J. Yu, and J. Zhang. "Zero-Shot Image Restoration Using Denoising Diffusion Null-Space Model". In: *The Eleventh International Conference on Learning Representations*. Sept. 2022. (Visited on 07/12/2026).